

Context

Tech Arena Italy 2026 is a Call for Ideas dedicated to excellent future talents from Italy and across Europe, strengthen university ties by connecting students/professors with Huawei through competitive tech challenges and community building.

SHAPE THE FUTURE OF **MEDIA TECHNOLOGY**

The Huawei Tech Arena is an open Call for Ideas aimed at talented university students, PhD candidates, and post-doctoral researchers. The initiative seeks to foster innovation, build a community of future Huawei talents and researchers, enable learning from industry experts, and offer the chance to compete for exciting prizes and internship opportunities. The top finalists will be invited to the final event at Huawei Italy R&D headquarters in Milan, where they will pitch their ideas to a jury of Huawei executives who will select the winners!

GENERAL TOPIC: NEXT-GENERATION RF TECHNOLOGY FOR WIRELESS COMMUNICATION

You can choose between these three challenges:



SUB-TOPIC 1: Up-Link enhancement techniques for Wireless Communication Systems

Uplink communication is increasingly critical for applications such as IoT, live streaming, and edge AI, yet remains constrained by UE power limits, interference, and hardware trade-offs. Emerging solutions such as reconfigurable antennas, advanced MIMO architectures, AI-driven optimization, and energy-efficient designs aim to improve coverage, increase uplink throughput, and enhance cell-edge performance in future 5G-Advanced and 6G networks.



SUB-TOPIC 2: Photonics for Wireless Communication Systems

Photonics-based wireless technologies such as Radio over Fiber (RoF) and Free Space Optical (FSO) communications enable ultra-high data rates, low latency, and interference-free transmission for future networks. Key challenges include signal distortion, weather sensitivity, integration complexity, power consumption, and scalability, driving the need for intelligent, robust, and energy-efficient photonic solutions.



SUB-TOPIC 3: Applications of AI to the modelling of EM problems

AI and neural networks are transforming electromagnetic design by enabling faster optimization, inverse design, and physics-informed modelling of complex EM structures. Key challenges include data availability, training cost, computational complexity, and ensuring Maxwell-consistent solutions while maintaining simulation-level accuracy and manufacturability.

Note: NN training datasets can be provided by Huawei upon request from participants."

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SUB-TOPIC 1: Up-Link enhancement techniques for Wireless Communication Systems

Technical Background

- Modern mobile networks (4G LTE, 5G NR, and emerging 6G) exhibit asymmetric traffic, with downlink historically prioritized. However, the rise of uplink-intensive applications—such as IoT, live streaming, and edge AI—has made uplink (UL) performance a critical bottleneck.
- Uplink communication is constrained by limited UE transmit power, higher interference, and challenging propagation conditions. While technologies like Massive MIMO and beamforming improve efficiency, they are largely downlink-optimized. Emerging reconfigurable and intelligent antenna systems offer new opportunities to enhance uplink reception, increase SNR, and improve cell-edge performance.

Technical Challenges

1. Power Limited Uplink - UE devices operate under strict power constraints, limiting link budget especially at higher frequencies (e.g., U6G and mmWave).
2. Hardware Limitations - Static antenna radiation patterns, Limited adaptability to frequency bands and traffic conditions, Trade-offs between gain, coverage, and cost
3. Interference Management - Inter-cell interference in dense deployments, Pilot contamination in massive MIMO systems, Limited spatial filtering flexibility in fixed antenna

Proposal: Up-link enhancement for future wireless communication systems

The proposed project is expected to provide innovative solutions to improve the Up-Link of mobile communication systems. The proposal could consider (but not limited to) the following directions:

1. Reconfigurable antennas with dynamic control of radiation performance.
2. Novel low-loss switching/tunable materials enabling low-cost reconfigurability
3. Enhanced MIMO architectures optimized for Up-Link

Potential innovation Proposals

1. Material-antenna co-design for optimized performance.
2. Hybrid electronic and material-based reconfiguration.
3. AI-driven self-optimizing antenna systems.
4. Modular and scalable reconfigurable MIMO framework.
5. Joint optimization across layers
6. Energy-efficient techniques to lower overall network energy consumption

Expected outcomes and Relevant KPIs

Proposed solutions should demonstrate benefits mainly in terms of:

1. improved uplink coverage and cell-edge performance;
2. increased uplink throughput and spectral efficiency

by assuming practical design constraints such as UE power-limited operation, implementation complexity, compatibility with 5G-Advanced/6G evolution.

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SUB-TOPIC 2: Photonics for Wireless Communication Systems

Technical Background

- Next-generation wireless networks such as 5G/6G, IoT, and cloud-based services require ultra-high data rates, low latency, and high reliability. Photonics-based technologies, particularly Radio over Fiber (RoF) and Free Space Optical (FSO) communications, enable high-bandwidth, low-loss, and interference-free signal transmission. Advances in photonic integration and optoelectronic components have improved system performance, but further development is needed for large-scale deployment.

Technical Challenges

Despite their advantages, photonics-based wireless communication systems face several technical challenges:

1. Signal distortion and nonlinear effects in RoF links.
2. Sensitivity of FSO systems to weather, turbulence, and misalignment.
3. Complex and costly integration of photonic and electronic components.
4. High power consumption and thermal management issues.
5. Limited adaptability to changing channel and network conditions.
6. Scalability and synchronization challenges in distributed systems.

Proposal: Develop intelligent, robust, and energy-efficient photonics-based wireless communication systems

The proposed project is expected to provide innovative solutions for photonics-based wireless communications.

The proposal could consider (but not limited to) the following directions:

1. Design high-performance RoF and FSO subsystems for future networks.
2. Develop low-noise, wideband photonic components either in silicon photonics or hybrid integration platforms (e.g. photonic oscillator for the generation of low-noise mmWave signals).
3. Implementation of Low-distortion optical modulation and demodulation schemes as well as dispersion and nonlinearity mitigation techniques for RoF transmission.
4. Enhance FSO link robustness under adverse conditions, by means of innovative adaptive optics, beam-steering and diversity techniques.
5. Implement adaptive and energy-efficient control mechanisms.

Potential innovation Proposals

1. Novel technologies and algorithms for turbulence compensation in FSO links.
2. Optical/Photonic beamforming technologies.
3. Hybrid MW/mm Wave-FSO transmission architecture or sub-systems.
4. Integrated terahertz photonics for ultra-high-speed links.
5. AI-enabled self-optimizing photonic networks.

Expected outcomes and relevant KPIs

Proposed solutions should demonstrate benefits mainly in terms of:

1. maximization of link length, link availability, optical bandwidth and optical power for FSO systems;
2. minimization of phase noise, NF and power consumption; maximization of Gain and linearity for RoF systems

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SUB-TOPIC 3: Applications of AI to the modelling of EM problems

Technical Background

- The recent development of Artificial Intelligence has enabled new methods to address the synthesis of specific EM structures using Neural Networks, opening the door to a larger set of technical solutions, which would not be possible by the simple application of rigorous engineering principles. These solutions often perform better than conventional solutions, or are more easily manufacturable or present more robustness towards manufacturing tolerance.

Technical Challenges

Even though the application of AI and Neural Networks shows promising potential, several practical issues still need to be addressed:

1. The availability of suitable data sets for the training of the Neural Network
2. The amount of data required for training the Neural Network, and the time required for the training process
3. Constrain the AI to generate Maxwell Equations-informed solutions
4. The high computational cost associated to AI-driven optimization of EM structures

AI- Driven Electromagnetic Design & Validation: Explore how Neural Networks can accelerate EM design

Combine deep learning with physics- based modeling to create faster, smarter electromagnetic design workflows.

1. Design your EM structure

- RF/Mw domain: high- gain antenna, RF circuit block, passive components
- Optical domain: a metalens or a photonic device.

2. Develop Physics- Informed Neural Networks (PINNs)

- Integrate Maxwell's equations directly into the learning model.
- Minimize training data while maintaining simulation- level accuracy

3. Validate and analyze

- Run sensitivity and tolerance analyses to evaluate geometry, material, and fabrication variations.
- Hundreds or thousands of simulations, infeasible with standard EM solvers.
- Use PINN surrogate models to perform these large parameter sweeps efficiently, enabling rapid, data- driven design exploration.

Some Innovation Proposals

1. **AI- accelerated design of broadband**, miniaturized antennas using PINN- based surrogates for multi- band optimization.
2. **Physics- informed neural networks for optical metasurfaces**, enabling ultra- fast shape tuning and phase control with minimal training data.
3. **Surrogate- based EM digital twins** to predict fabrication tolerance impacts across thousands of design variants in seconds.
4. **AI- assisted inverse design frameworks** where desired EM responses directly generate optimized geometries.

Expected Outcomes and Relevant KPIs

1. **AI- accelerated design**, outlining gain in terms of time, computing resources against conventional engineering design methods
2. **PINN-based Sensitivity analysis of AI- accelerated design**, outline gain in terms of time, computing resources against conventional Monte Carlo Analysis

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